

IoT & Embedded Systems Security

What Makes IoT Different

IoT devices – smart lights, cameras, thermostats, industrial sensors – are computers with three dangerous properties:

They are often unmanaged – no one patches them

They run full TCP/IP stacks – directly reachable on networks

They are physically deployable – attackers can buy them, tear them apart, and extract secrets

Embedded Hardware Components

Component	Role
Microcontroller (MCU)	Single chip computer (ARM Cortex-M, ESP32, etc.)
SoC	System-on-Chip (attacker router, camera)
Flash/EEPROM	Holds firmware
UART/JTAG	Debug interfaces (attackers love unauthenticated serial)
EMI shielding & OTP fuses	Physical protection against probing

Firmware Is the Operating System

Embedded firmware is usually a monolithic blob. Attackers:

Download the firmware image from the vendor

Use firmware analysis tools to extract the file system (`binwalk`, `strings`, `ghidra`)

Find hardcoded credentials, backdoors, and vulnerabilities

The infamous Mirai botnet (2016) did not use sophisticated exploits. It used ~60 default usernames/passwords. Every device in the camera/DVR market had the same defaults – easy pickings for a botnet that nearly took down the internet's DNS infrastructure.

The IoT Threat Model

Device compromise: default creds, unpatched CVEs, exposed debug ports

Network exposure: devices on the same LAN as your laptop

Data privacy: cameras and mics streaming to remote vendors

Physical tampering: JTAG/SWD on a smart lock

DoS/Vendors: permanent device control, bricking

Securing IoT (Defensive Checklist)

Change every default password

Segment IoT on an isolated VLAN/guest network

Disable UPnP and remote management

Block IoT device outbound connections except required

Check for firmware updates monthly

Block vendor phone-home domains at the firewall

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